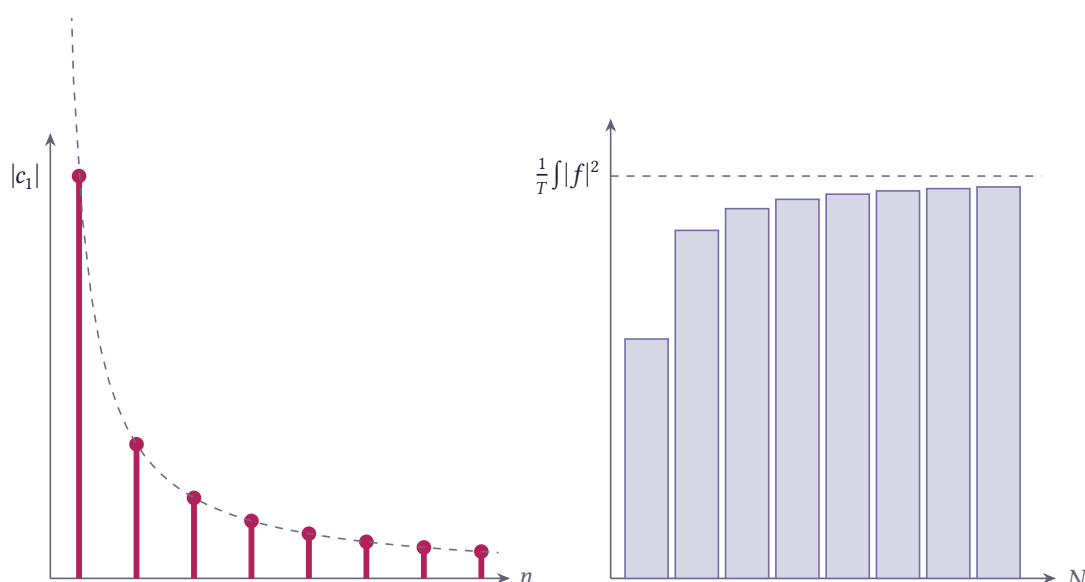


Properties of Fourier Series, Parseval's Theorem and the Complex Form

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In this chapter

9.1 Approximating a Periodic Function by a Finite Trigonometric Series	2
9.2 Differentiation and Integration of Fourier Series	3
9.3 Rayleigh's Theorem	3
9.4 Parseval's Theorem	4
9.5 The Complex Form of the Fourier Series	5

9.1 Approximating a Periodic Function by a Finite Trigonometric Series

Theorem 9.1

For an N -term approximation of a periodic function $f(x)$ by a series of sines and cosines, the best coefficients (in the MMSE sense) are the Fourier series coefficients.

Consider a periodic function with period 2π :

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx, \quad f(x) = f(x + 2\pi)$$

The N -term trigonometric approximation:

$$f(x) \approx a_0 + \sum_{n=1}^N a_n \cos nx + b_n \sin nx$$

Is the following trigonometric approximation not **better**?

$$F(x) \triangleq \alpha_0 + \sum_{n=1}^N \alpha_n \cos nx + \beta_n \sin nx$$

What is the **criterion** for an approximation being better? One criterion: the error function:

$$E(x) = |f(x) - F(x)| = \left| f(x) - \alpha_0 - \sum_{n=1}^N (\alpha_n \cos nx + \beta_n \sin nx) \right|$$

A good criterion: the mean squared error:

$$D \triangleq \int_{-\pi}^{\pi} E^2(x) dx = \int_{-\pi}^{\pi} \left| f(x) - \alpha_0 - \sum_{n=1}^N (\alpha_n \cos nx + \beta_n \sin nx) \right|^2 dx$$

Proof.

For the best approximation we minimize the mean squared error. This is done by differentiating D with respect to the coefficients of the trigonometric series and setting these derivatives equal to zero. In other words:

$$\begin{aligned} \frac{\partial D}{\partial \alpha_0} &= -2 \int_{-\pi}^{\pi} \left[f(x) - \alpha_0 - \sum_{n=1}^N (\alpha_n \cos nx + \beta_n \sin nx) \right] dx = 0 \\ \int_{-\pi}^{\pi} f(x) dx &= \int_{-\pi}^{\pi} \left[\alpha_0 + \sum_{n=1}^N (\alpha_n \cos nx + \beta_n \sin nx) \right] dx = 2\pi \alpha_0 \implies \alpha_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx = a_0 \\ \frac{\partial D}{\partial \alpha_m} &= 0 \implies \alpha_m = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos mx dx = a_m \\ \frac{\partial D}{\partial \beta_m} &= 0 \implies \beta_m = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin mx dx = b_m \end{aligned}$$

The best finite trigonometric approximation of a periodic function is the finite Fourier series. ■

9.2 Differentiation and Integration of Fourier Series

Theorem 9.2 ■ Differentiating a Fourier series

If $f(x)$ is periodic with period T , continuous and has a Fourier series, then its derivative is also periodic, and its Fourier series is obtained by differentiating the Fourier series of the function.

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos \frac{2\pi}{T} nx + b_n \sin \frac{2\pi}{T} nx$$

$$\Rightarrow f'(x) = \sum_{n=1}^{\infty} \left(\frac{2\pi}{T} n b_n \right) \cos \frac{2\pi}{T} nx + \left(-\frac{2\pi}{T} n a_n \right) \sin \frac{2\pi}{T} nx$$

Theorem 9.3 ■ Integrating a Fourier series

If $f(x)$ is periodic with period T and has a Fourier series, then the definite integral of this function can be computed by integrating the Fourier series of the function.

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos \frac{2\pi}{T} nx + b_n \sin \frac{2\pi}{T} nx$$

$$\Rightarrow \int_0^x f(t) dt = a_0 x + \sum_{n=1}^{\infty} \left(\frac{T}{2\pi n} a_n \sin \frac{2\pi}{T} nx - \frac{T}{2\pi n} b_n \cos \frac{2\pi}{T} nx + \frac{T}{2\pi n} b_n \right)$$

9.3 Rayleigh's Theorem

Theorem 9.4 ■ Rayleigh's theorem

If the functions $f(x)$ and $g(x)$ are periodic with period T and their Fourier series exist as

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos \frac{2\pi}{T} nx + b_n \sin \frac{2\pi}{T} nx$$

$$g(x) = \alpha_0 + \sum_{n=1}^{\infty} \alpha_n \cos \frac{2\pi}{T} nx + \beta_n \sin \frac{2\pi}{T} nx$$

then

$$\frac{2}{T} \int_T f(x) \overline{g(x)} dx = 2a_0 \overline{\alpha_0} + \sum_{n=1}^{\infty} a_n \overline{\alpha_n} + b_n \overline{\beta_n}$$

Proof.

Proof, assuming $g(x)$ is real:

$$\begin{aligned}
 \frac{2}{T} \int_T f(x)g(x) dx &= \frac{2}{T} \int_T f(x) \left(\alpha_0 + \sum_{n=1}^{\infty} \alpha_n \cos \frac{2\pi}{T} nx + \beta_n \sin \frac{2\pi}{T} nx \right) dx \\
 &= \frac{2}{T} \int_T f(x) \alpha_0 dx + \sum_{n=1}^{\infty} \frac{2}{T} \int_T f(x) \left(\alpha_n \cos \frac{2\pi}{T} nx + \beta_n \sin \frac{2\pi}{T} nx \right) dx \\
 &= \underbrace{\frac{2}{T} \int_T f(x) dx}_{2a_0} \alpha_0 + \sum_{n=1}^{\infty} \underbrace{\frac{2}{T} \int_T f(x) \cos \frac{2\pi}{T} nx dx}_{a_n} \alpha_n + \underbrace{\frac{2}{T} \int_T f(x) \sin \frac{2\pi}{T} nx dx}_{b_n} \beta_n \\
 &= 2a_0 \alpha_0 + \sum_{n=1}^{\infty} a_n \alpha_n + b_n \beta_n
 \end{aligned}$$

■

9.4 Parseval's Theorem

Theorem 9.5 ■ Parseval's theorem

If the function $f(x)$ is periodic with period T and its Fourier series exists as

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos \frac{2\pi}{T} nx + b_n \sin \frac{2\pi}{T} nx$$

then

$$\frac{2}{T} \int_T |f(x)|^2 dx = 2|a_0|^2 + \sum_{n=1}^{\infty} (|a_n|^2 + |b_n|^2)$$

Proof.

In Rayleigh's identity, replacing $g(x)$ by $f(x)$ gives Parseval's identity.

■

By Parseval's identity, the power of the function $f(x)$ can be obtained from the Fourier coefficients. If the function is expressed by an N -term approximation of sines and cosines, then by the relation above the power of the approximating function is less than the power of $f(x)$, and we have:

Key result 9.6 ■ Bessel's inequality

$$\frac{2}{T} \int_T |f(x)|^2 dx \geq 2|a_0|^2 + \sum_{n=1}^N (|a_n|^2 + |b_n|^2)$$

Example 9.7

For the function $f(x)$ below, how many terms of the Fourier series are needed for at least 94 percent of the power to remain in the approximating function?

$$f(x) = \sum_{n=1}^{\infty} \frac{2k}{n\pi} (1 - \cos n\pi) \sin nx$$

$$\frac{2}{T} \int_T |f(x)|^2 dx = \frac{1}{\pi} \int_{-\pi}^{\pi} k^2 dx = \frac{2}{\pi} [k^2 x]_0^{\pi} = 2k^2$$

$$\frac{\sum_{n=1}^N \left(\frac{2k}{n\pi} (1 - \cos n\pi) \right)^2}{2k^2} \geq 0.94$$

$$\frac{\sum_{n=1}^7 \left(\frac{2k}{n\pi} (1 - \cos n\pi) \right)^2}{2k^2} \cong 0.9496, \quad \frac{\sum_{n=1}^5 \left(\frac{2k}{n\pi} (1 - \cos n\pi) \right)^2}{2k^2} \cong 0.9331 \implies N = 7$$

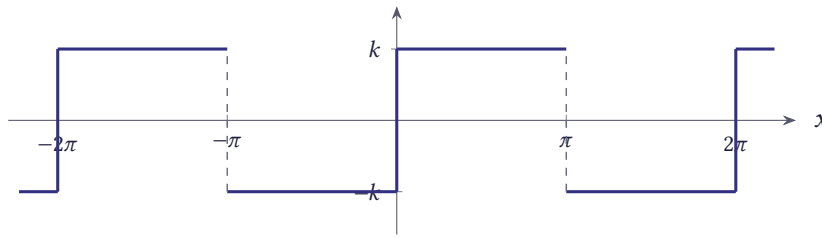


Figure 9.1 The square wave $\pm k$ with period 2π used in the power example

9.5 The Complex Form of the Fourier Series

Suppose $f(x)$ is periodic and, merely for simplicity, has period 2π . Then

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx$$

Using Euler's formula $e^{ix} = \cos x + i \sin x$:

$$\cos nx = \frac{e^{inx} + e^{-inx}}{2}, \quad \sin nx = \frac{e^{inx} - e^{-inx}}{2i}, \quad \frac{1}{i} = -i$$

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \left(\frac{e^{inx} + e^{-inx}}{2} \right) + b_n \left(\frac{e^{inx} - e^{-inx}}{2i} \right) = a_0 + \sum_{n=1}^{\infty} \underbrace{\left(\frac{a_n - ib_n}{2} \right)}_{c_n} e^{inx} + \underbrace{\left(\frac{a_n + ib_n}{2} \right)}_{c_{-n}} e^{-inx}$$

$$c_n = \frac{a_n - ib_n}{2} = \frac{1}{2} \left\{ \frac{1}{\pi} \int_{2\pi} f(x) \cos nx dx - i \frac{1}{\pi} \int_{2\pi} f(x) \sin nx dx \right\}$$

$$= \frac{1}{2\pi} \int_{2\pi} f(x) (\cos nx - i \sin nx) dx = \frac{1}{2\pi} \int_{2\pi} f(x) e^{-inx} dx$$

$$c_{-n} = \frac{a_n + ib_n}{2} = \frac{1}{2\pi} \int_{2\pi} f(x)(\cos nx + i \sin nx) dx = \frac{1}{2\pi} \int_{2\pi} f(x)e^{inx} dx$$

Key result 9.8 ■ The complex form of the Fourier series

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{inx}, \quad c_n = \frac{1}{2\pi} \int_{2\pi} f(x)e^{-inx} dx$$

For an arbitrary period:

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{i\frac{2\pi}{T}nx}, \quad c_n = \frac{1}{T} \int_T f(x)e^{-i\frac{2\pi}{T}nx} dx$$

$$\frac{1}{T} \int_T |f(x)|^2 dx = \sum_{n=-\infty}^{\infty} |c_n|^2$$